

Mohammed Shawqi Aftab

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Address: 216 Anwar Pasha Bhaban, Secretariat Road, West Fullbaria, Dhaka

RESEARCH INTERESTS

Parameterized Algorithms, Formal Verification & SMT Solving, Graph Theory & Structural Graph Parameters, Algorithm Design & Analysis, Computational Complexity, Data Structures & Online Algorithms

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science and Engineering

East West University

Feb. 2022 – Dec. 2025

- CGPA: 4.00/4.00 (*Top 0.29% – 1st out of 348 CSE graduates*)
- **Summa Cum Laude · Gold Medalist**, Convocation 2026

High School Certificate (HSC.)

Dhaka College

2020

- GPA: 5.00/5.00

Secondary School Certificate (SSC.)

Udayan Uchcha Madhyamik Bidayala

2018

- GPA: 5.00/5.00

TEACHING EXPERIENCE

Teaching Assistant

Semesters: Spring 2025, Summer 2025, Fall 2025

Department of CSE, East West University

- Assisted in conducting courses: CSE 106 - Discrete Mathematics, CSE 207 - Data Structure, CSE 246- Algorithm, CSE 407 - Green Computing, CSE 302 - Database Systems, CSE 303 - Statistics for Data Science, CSE 488 - Big Data Analysis
- Conducted tutorial sessions and lab classes and helped student understand teaching materials for 30 - 40 students per section depending on course
- Evaluated assignments, quizzes, and examination scripts
- Provided academic guidance

Competitive Programming Trainer

East West University

- Trained students in algorithmic problem solving and competitive programming

COMPETITIVE PROGRAMMING BACKGROUND

Online Judge Profiles

- **Codeforces:** [Dr.KeK](#)
- **CodeChef:** [drkek](#)
- **AtCoder:** [DrKeK](#)

Notable Contest Achievements

- **5th Place** - BUBT Inter University Collaborative Programming Contest
- **6th Place** - Uttara University Inter-University Programming Contest 2025
- **9th Place** - DUET Inter University Programming Contest (IUPC) 2025
- **12th Place** - 8th DRMC International Tech Carnival 2025 - Programming Contest - Main Round
- **12th Place in Bangladesh, 1049th Globally** - Meta Hacker Cup Round 2, 2025
- **30th Place** - ICPC Preliminary Dhaka Site 2024
- **39th Place** - BUET Inter University Programming Contest 2024
- **39th Place** - KUET Inter University Programming Contest 2024
- **48th Place** - ICPC Dhaka Regional Site 2025

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, SQL
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy
- **Web Technologies:** HTML, CSS, React, Node.js, Django, Flask
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Technologies:** Git, Linux, LaTeX, MATLAB
- **Areas of Expertise:** Algorithm Design, Machine Learning, Deep Learning, Data Analysis

PUBLICATIONS

1. **Mohammed Shawqi Aftab**, Rakib Hossen, Jia Uddin, and Jasim Uddin. “Unified Detection Framework for SQL Injection and Cross-Site Scripting Using Multi-Architecture Comparison.” *PLOS ONE*. **Under review; second-round revision, 2026.**
2. Jerin Anan Proma, Fatema Tuz Zannat, **Mohammed Shawqi Aftab**, and Shamim H. Ripon. “Graph-based EEG Emotion Recognition with Facial Video Fusion on DEAP: Within-Subject Gains and Cross-Subject Limits.” *PLOS ONE*. **Under review, 2026.**
3. Jerin Anan Proma, Fatema Tuz Zannat, **Mohammed Shawqi Aftab**, Tahura Akter Tripty, M. Saddam Hossain Khan, Raihan Ul Islam, and Rashedul Amin Tuhin. “Energy-Efficient Web Design: Measuring Impact of Front-end Optimizations.” In: *Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS '25)*, pp. 92–100. ACM, 2025. DOI: [10.1145/3777555.3777561](https://doi.org/10.1145/3777555.3777561).
4. Sourav Hasan, Nushera Tajrin Mimou, **Mohammed Shawqi Aftab**, B. M. Shahria Alam, Golam Kibria, Md. Parvezur Rahman Mahin, Rafsun Islam Taskin, and Ahmed Wasif Reza. “Potential Implications of Automated Detection and Grad-CAM Visualization on Ishihara Test Plates.” In: *Data Mining and Information Security (ICDMIS 2025)*, Lecture Notes in Networks and Systems, vol. 1972, pp. 315–331. Springer, 2026.

PROFESSIONAL ACTIVITIES

- **Peer Reviewer:** PLOS ONE (2026)

AWARDS & HONORS

- **Summa Cum Laude and Gold Medal**, East West University Convocation 2025 — Highest CGPA among all CSE graduates (4.00/4.00; 1st of 348)
- **Summa Cum Laude**, East West University Convocation 2025
- **Merit Scholarship**, East West University, Spring 2025 (CGPA: 4.00)
- **Merit Scholarship**, East West University, Spring 2024 (CGPA: 4.00)
- **Merit Scholarship**, East West University, Spring 2023 (CGPA: 4.00)
- **Dean’s List**, East West University, Multiple Semesters (2022–2025)

- **Art Accepted**, Charcoal Art Exhibition, BUET Artista Society, 2023

LANGUAGES

- **Bengali:** Native
- **English:** Fluent